Tanmay Pandey

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Short Biography

I am Tanmay Pandey, a fourth year Biology majors, studying in **Indian Institute of Science Education and Research, Mohali**, India. I am interested in Membrane Biophysics, Synthetic Biology, Neuromorphic Computation, and Cell Signalling Biophysics.

Publications

Experimentally Determined Shapes of Plasma Membrane Vesicles, Phosphatidylcholine (PC), and PC-Cholesterol Vesicles: Vesicles Deflation Analysis Using Confocal Microscopy — Accepted - 29/05/2025
Journal of Physical Chemistry B
Harshmeet Kaur, Rajni Kudawla, ***Tanmay Pandey***, Tripta Bhatia
Preprint available on bioRxiv

Shape analysis of Biomimetic and Plasma Membrane Vesicles — October 2024
ChemSystemsChem
Rajni Kudawla, Harshmeet Kaur, ***Tanmay Pandey***, Tripta Bhatia
DOI: 10.1002/syst.202400052

Education

BS-MS **Indian Institute of Science Education and Research, Mohali**, India — October 2022 – July 2027

- Biology Majors, and Physics Minor
- CPI : 7.54/10

Projects

DNA-based neuromorphic Computation — Remote, March 2024 – Ongoing
Bio-inspired computation lab, *University of Kiel*
Supervisor: Prof. Dr. Jan Steinkuelher

Shape analysis of biomimetic and plasma membrane vesicles — IISER, Mohali, June 2023 – March 2025
Soft matter Biophysics Lab, *IISER Mohali*
Supervisor: Dr. Tripta Bhatia

Algorithms for the analysis of micropipette aspiration microscopy data — IISER, Mohali, June 2023 – August 2023
Soft matter Biophysics Lab, *IISER Mohali*
Supervisor: Dr. Tripta Bhatia

Other Courses

Lipid Membranes: From Cells to Synthetic Biology, by *Dr. Tripta Bhatia* and *Dr. Thomas G. Pomorski*. — Global Inititative of Academic Networks

Skills

Programming Languages: Python, Javascript, Matlab

Laboratory Skills: GUV preparation, Confocal microscopy, Phase contrast microscopy, Image analysis using OpenCV and ImageJ